Concept 1 - Factors

2, 3, 4, 6, 12 }

$\frac{12}{2}$ ✓ $\frac{12}{3}$ ✓ $\frac{12}{4}$ ✓ $\frac{12}{6}$ ✓ $\frac{12}{12}$ ✓

$\frac{12}{\ }$ ×

$12 = a \times b$ [a & b are factors of 12]

$12 = 1 \times 12$

$= 2 \times 6$

$\times 4$

[1, 2, 3, 4, 6, 12 are factors of 12]

$a \times b = c$

a, b are factors of c

$\frac{c}{a} = b$ ✓

$\frac{c}{b} = a$ ✓

Number System

Number of Factors

Prime Factorization

M1

2	12
2	6
3	3
	1

$12 = 2 \times 2 \times 3$

$\boxed{12 = 2^2 \times 3^1}$

Number of factors: $(2+1)(1+1)$

$: 3 \times 2$

$: 6$

M2

$12 = 4 \times 3 = 2^2 \times 3$

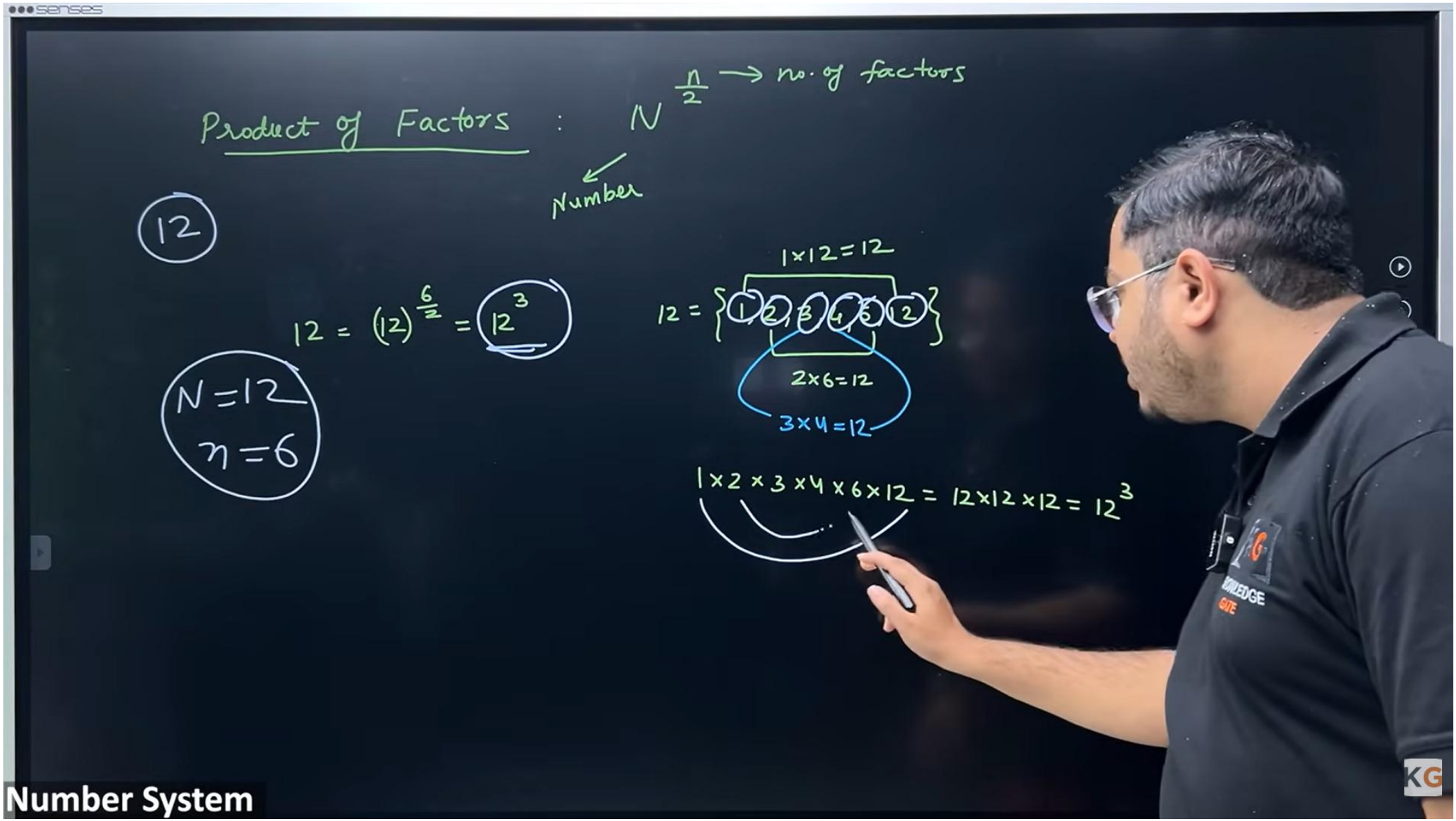
$12 = 2 \times 6 = 2 \times 2 \times 3 = 2^2 \times 3$

$N = p^a \times q^b \times r^c$

$p, q, r \rightarrow$ prime numbers

Number of factors:

$(a+1)(b+1)(c+1)$



Number System

Q1: Find Number of Factors and Product of Factors of 72.

2	72
2	36
2	18
3	9
3	3
	1

$72 = 2 \times 2 \times 2 \times 3 \times 3$

$= 2^3 \times 3^2$

$72 = 9 \times 8$

$= 3^2 \times 2^3$

$n = (3+1)(2+1)$

$= 4 \times 3 = 12$

1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72

$1 \times 72 = 72$
$2 \times 36 = 72$
$3 \times 24 = 72$
$4 \times 18 = 72$
$6 \times 12 = 72$
$8 \times 9 = 72$

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KNOWLEDGE GATE

Number System

Q2: Consider the expression: $(999....9)^2$, if there are a total of 2020 9's, then find the following:

a) Total number of digits in the expression

b) Value of the expression

c) Digits Sum

(b) $(\underbrace{9999\ldots9}_{2020})^2 = \underbrace{9999\ldots9}_{2019}\,8\,\underbrace{00000\ldots0}_{2019}\,1$

(c) $\underbrace{9+9+9+\ldots9}+8+0+0+\ldots+0+1$

$(2019\times9)+8+1 = (2019\times9)+9 = 2020\times9 = 18{,}180$

(a) Total digits

$2019 + 1$

$+\ 2019 + 1$

$2020 + 2020$

4040

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Q3: Consider the expression: $(999....9)^3$, if there are a total of 2020 9's, then find the following:

a) Total number of digits in the expression

b) Value of the expression

c) Digits Sum

(b) $\underbrace{(9999..9)}_{2020}^{3} = \underbrace{999...9}_{2019}\,7\,\underbrace{000..0}_{2019}\,2\,\underbrace{9999..9}_{2020}$

(c) $\underbrace{9+9+9+...+9}_{2019}+\underbrace{7+0+0+...0+2}_{1}+\underbrace{9+9+9+...+9}_{2020}$

$4040 \times 9 = 36,360$

(a) Total digits

2019
+ 1
+ 2019
+ 1
+ 202

2
+

Number System

Q4. The square root of 12345678987654321 is nnnnnn.... upto 'p' times, find the sum of n and p ?

$\sqrt{12345678987654321} = \underbrace{nnnn\ldots}_{p \text{ times}}$

11111...

$n+p=??$

$p=9 \quad n=1$

$n+p=10$

✓ $(11111)^2$

× $(9999..)^2$

9...

$(nnnn...)^2 = 12345678987654321$

111 ①

2222 ④

3337 ⑨

444 ⑥

555 ⑤

666 ⑥

777^2 ⑨

888^2 ④

999 ①

Q5. A number is divided by 406 leaves remainder 115 , What will be the reminder when it will be divided by 29?

M2

$406 \overline{)N}$, remainder 115

$\Rightarrow$ $406 \overline{)N}$ = 1, -406, remainder 115

$\Rightarrow N - 406 = 115$

$N = 115 + 406 = 521$

$3 \overline{)25}$ = 8, -24, remainder 1

$\Rightarrow \frac{N}{29} = \frac{521}{29} \Rightarrow R(28)$

$29 \overline{)521}$ = 17, -29, 231, -203, remainder 28

Trick

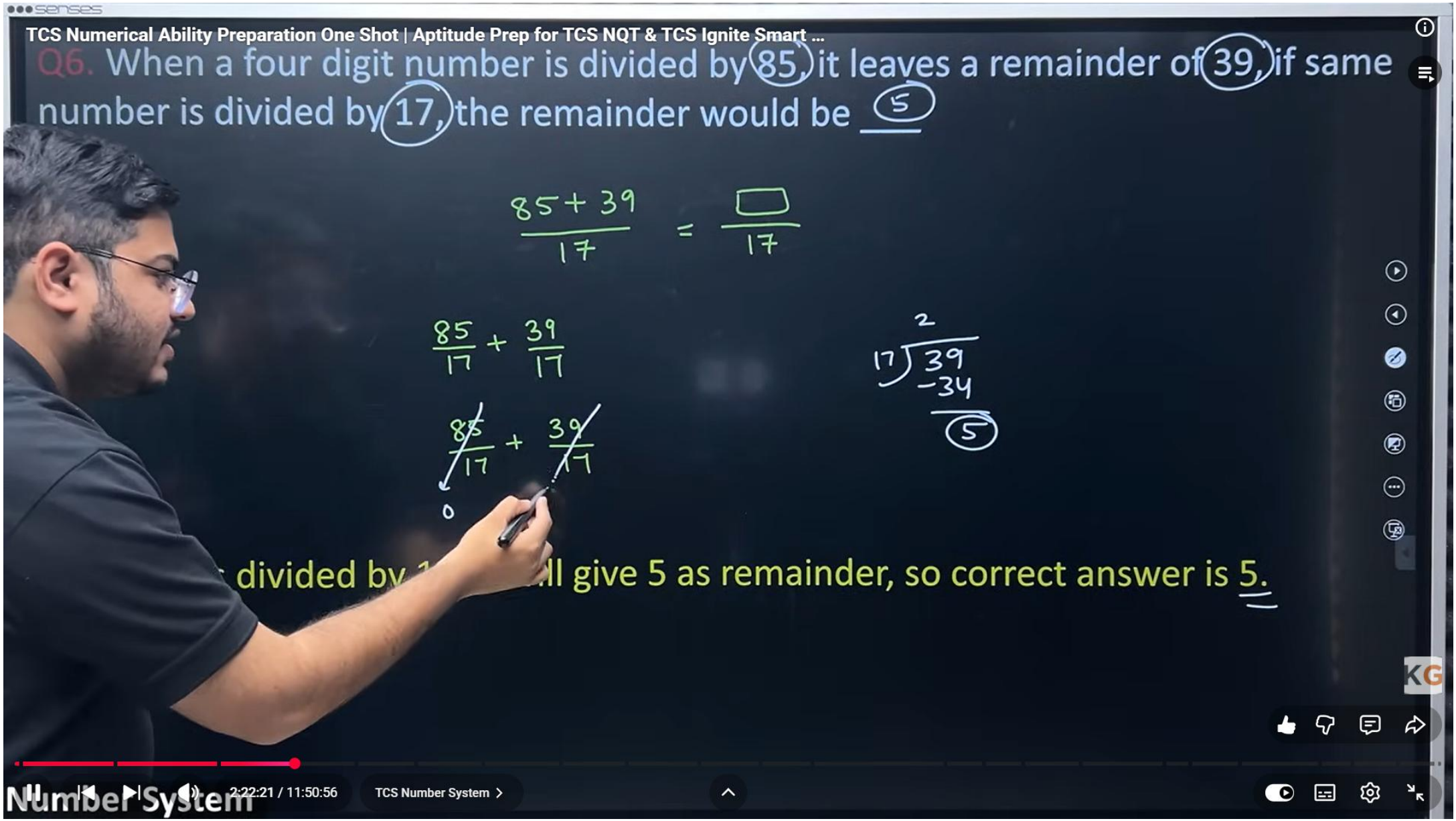
$406 \overline{)N}$, remainder 115

$\frac{406+115}{29} \Rightarrow \frac{\text{Divisor + Given Rem}}{\text{Target Rem}}$

$\underbrace{\frac{521}{29}}_{28}$ or $\frac{406+115}{29} = \underbrace{\frac{406}{29}}_{0} + \underbrace{\frac{115}{29}}_{28}$

$= 28$

Number System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Q6. When a four digit number is divided by 85, it leaves a remainder of 39, if same number is divided by 17, the remainder would be 5
85 + 39 / 17 = □ / 17
85/17 + 39/17
85/17 + 39/17
0
2
17) 39
−34
5
divided by 1 ll give 5 as remainder, so correct answer is 5.
2:22:21 / 11:50:56
TCS Number System
Number System

senses
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Correct Answer : Option D

Explanation -

Ans: Both have the same value

For packet A : 1 gm cost = 72/100 =Rs. 18/25

For Packet B : 1 gm cost = 108/150 =Rs. 18/25

Thus both packets have the same value.

KG
Arithmetic
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TCS Arithmetic

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Q3. 6 cows cost the same as 9 sheep, 27 sheep cost the same as 30 goats. 50 goats cost the same as 3 elephants. If two elephants cost Rs 7200, then the cost of one cow is

1 cow → C
1 sheep → S
1 goat → G
1 elephant → E

2 elephants ⇒ 2E

$2E = 7200$

$E = \frac{7200}{2} = 3600$

$50G = 3E$

$1G = \frac{3E}{50}$

$30G = \frac{3E}{\cancel{50}\,5} \times \cancel{30}^{3}$

$30G = \frac{9E}{5}$

$\boxed{27S \quad 30G}$

$6C = 9S$

$\downarrow \times 3 \qquad \downarrow \times 3$

$18C = 27S$

Arithmetic
2:42:04 / 11:50:56
TCS Arithmetic

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Correct Answer : Option A

Explanation –

Let the cost of each item be Y, then X = 9600/Y =>

XY=9600 —(i)

When cost is reduced 20, we have (X+40) = 9600 / (Y-20)

=> XY -20X + 40Y - 800 = 9600

=> 9600 - 20X + 40Y - 800 = 9600

=> -20X + 40Y = 800 —(ii)

As we require X, so substituting Y=9600/X in (ii)

=> -20X + 40*(9600/X) = 800

=> XÂ² + 40X - 19200 = 0

=> (X-120) (X+160) = 0

Considering +ve value for X, we have X=120.

Arithmetic
2:48:11 / 11:50:56
TCS Arithmetic